Kernel Methods for the Approximation of the Eigenfunctions of the Koopman Operator

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Abstract

The Koopman operator provides a linear framework to study nonlinear dynamical systems. Its spectra offer valuable insights into system dynamics, but the operator can exhibit both discrete and continuous spectra, complicating direct computations. In this paper, we introduce a kernel-based method to construct the principal eigenfunctions of the Koopman operator without explicitly computing the operator itself. These principal eigenfunctions are associated with the equilibrium dynamics, and their eigenvalues match those of the linearization of the nonlinear system at the equilibrium point. We exploit the structure of the principal eigenfunctions by decomposing them into linear and nonlinear components. The linear part corresponds to the left eigenvector of the system's linearization at the equilibrium, while the nonlinear part is obtained by solving a partial differential equation (PDE) using kernel methods. Our approach avoids common issues such as spectral pollution and spurious eigenvalues, which can arise in previous methods. We demonstrate the effectiveness of our algorithm through numerical examples.

1 Introduction

Time series data, which are ubiquitous in various scientific domains, have driven the development of a wide range of statistical and machine learning forecasting methods [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18]. Dynamical systems theory provides tools to understand the underlying rules governing the evolution of time series data. Originating with the work by Koopman [19], Koopman operator theory provides a linear lens to study nonlinear dynamical systems. In particular, the Koopman operator maps the nonlinear evolution of finite-dimensional state space to an infinite-dimensional but linear description of functions. Following the seminal work [20], there is a surge in interest in using the Koopman operator to study dynamical systems. As a linear operator, the spectral of the Koopman operator is rich in information and plays a significant role in various analysis and synthesis problems. For example, the principal eigenfunctions reveal the state space geometry as the joint zero-level curves characterize the stable and unstable manifolds of the equilibrium points [21, 22]. In [23], the principal eigenfunctions are used to identify the stability boundary, thereby determining the domain of attraction of an equilibrium point. Furthermore, the solution to the optimal control problem can also be constructed using the principal eigenfunctions of the Koopman operator, per [24, 25]. In [26], the authors proposed a path-integral formula for the computation of Koopman principal eigenfunctions. Yet in general, the Koopman operator can have both discrete and continuous spectra [20], making the computation challenging. In addition, approximating the spectra of an infinite-dimensional operator via a finite-dimensional matrix can suffer from "spectral pollution" [27]. In this paper, we aim to develop a numerical method to construct the principal eigenfunction of the Koopman operator directly from data that circumvents these difficulties.

Numerical methods such as extended dynamic mode decomposition (EDMD) [28] aim to approximate the infinite-dimensional Koopman operator via its actions on the finite-dimensional space spaced by a set of pre-selected functions. Recent research has also explored the combination of kernel methods and Koopman operator theory [29, 30, 31, 32, 33, 34]. Building upon the theory of reproducing kernel Hilbert spaces (RKHS) [35], kernel-based methods offer considerable advantages in regularization, guaranteed convergence, automatization, and interpretability [36, 37, 38, 39], and have notably strengthened the mathematical basis for analyzing dynamical systems [3, 40, 41, 42, 43, 44, 45, 29, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56] as well as surrogate modeling [57]. In this paper, we explore the use of kernel methods for the computation of spectral properties of the Koopman operator. The advantage of using kernel methods in this context is that they provide with error estimates that make the method more rigorous.

The existing literature on kernel-based Koopman learning, e.g. [29, 33], requires constructing a Koopman operator from data first. The spectra are then computed from the learned Koopman operator. Yet the spurious eigenvalues [58] can arise due to numerical artifacts, noise, or overfitting in the computation. These spurious eigenvalues do not represent genuine modes of the underlying dynamics and are misleading when analyzing the system's behavior. To deal with this challenge, we aim to directly learn the spectrum of the Koopman operator without computing the Koopman operator itself by solving a partial differential equation (PDE) in RKHS. Specifically, following [26], we associate the principal eigenfunctions of the Koopman operator with an equilibrium point with the corresponding eigenvalues being the same as that of the linearization of the nonlinear system at said equilibrium point. The linear parts of the principal eigenfunction correspond to the eigenvector of local linearization at the origin, while the nonlinear part of the eigenfunction is the solution of PDEs, which can be solved in RKHS. Unlike [59] that learns eigenfunctions of the Koopman operator in RKHS, the decomposition into linear and nonlinear parts allows us to incorporate knowledge from the linearization of system dynamics, and exploit the possible nonlinear structure of RKHS. Throughout the paper, we assume we have explicit access to the vector field $f(x)$ of each dynamical system we study.

The remainder of this paper is organized as follows: In Section 2, we provide background on the Koopman operator and its eigenfunctions. In Section 3, we discuss kernel methods and their application to solving the PDE satisfied by the Koopman eigenfunctions. Section 4 presents error estimates for our method, and in Section 5, we showcase numerical experiments.

2 Preliminaries on the Koopman Operator

2.1 Koopman Operator and its Spectrum

In this section, we provide a brief overview of the spectral theory of the Koopman operator. We refer the reader to [60, 61] for more details. Consider the dynamical system

$$\dot{x} = f(x), \tag{1}$$

defined on a state space $\mathcal{Z} \subseteq \mathbb{R}^p$. The vector field $\mathbf{f}$ is assumed to be smooth function. Let C^0 denote the space of continuous functions from $\mathcal{Z}$ to $\mathbb{C}$ and let $\mathcal{F} \subseteq C^0$ be the function space of observable $\psi : \mathcal{Z} \to \mathbb{C}$. We have following definitions for the Koopman operator and its spectrum.

Definition 1 (Koopman Operator)**.** *The family of Koopman operators $\mathbb{U}_t : \mathcal{F} \to \mathcal{F}$ corresponding to* (1) *is defined as*

$$[\mathbb{U}_t\psi](x) = \psi(s_t(x)). \tag{2}$$

where $s_t(x)$ is the solution of the dynamical system (1). If in addition ψ is continuously differentiable, then $\varphi(x, t) := [\mathbb{U}_t\psi](x)$ satisfies a partial differential equation

[62]

$$\frac{\partial \varphi}{\partial t} = \frac{\partial \varphi}{\partial x} f =: \mathcal{K}_f \varphi \tag{3}$$

with the initial condition $\varphi(x, 0) = \psi(x)$. The operator $\mathcal{K}_{\mathbf{f}}$ is the infinitesimal generator of $\mathbb{U}_t$ i.e.,

$$\mathcal{K}_f \psi = \lim_{t \to 0} \frac{(\mathbb{U}_t - I)\psi}{t}. \tag{4}$$

It is easy to check that each $\mathbb{U}_t$ is a linear operator on the space of functions, $\mathcal{F}$.

Definition 2. *[Eigenvalues and Eigenfunctions of Koopman] A function $\psi_\lambda(x)$, assumed to be at least C^1, is said to be an eigenfunction of the Koopman operator associated with eigenvalue λ if*

$$[\mathbb{U}_t \psi_\lambda](x) = e^{\lambda t} \psi_\lambda(x). \tag{5}$$

Using the Koopman generator, the (5) can be written as

$$\frac{\partial \psi_\lambda}{\partial x} f = \lambda \psi_\lambda. \tag{6}$$

The eigenfunctions and eigenvalues of the Koopman operator enjoy the following property [60, 63].

In this paper, we are interested in approximating the eigenfunctions of the Koopman operator with associated eigenvalues, the same as that of the linearization of the nonlinear system at the equilibrium point. With the hyperbolicity assumption on the equilibrium point of the system (1), this part of the spectrum of interest to us is well-defined. In the following discussion, we summarize the results from [60] relevant to this paper and justify some of the claims made above on the spectrum of the Koopman operator.

Equations (5) and (6) provide a general definition of the Koopman spectrum. However, the spectrum can be defined over finite time or over a subset of the state space. The spectrum of interest to us in this paper could be well-defined over the subset of the state space.

Definition 3 (Open Eigenfunction [60])**.** *Let $\psi_\lambda : \boldsymbol{\mathcal{C}} \to \mathbb{C}$, where $\boldsymbol{\mathcal{C}} \subset \mathcal{Z}$ is not an invariant set. Let $\tau^-(x)$ and $\tau^+(x)$ be the greatest lower and least upper bounds of the time interval on which the solution $x \in \boldsymbol{\mathcal{C}}$ exists, and $\tau \in (\tau^-(x), \tau^+(x)) = I_x$ be a connected open interval such that $\tau(x) \in \boldsymbol{\mathcal{C}}$ for all times $\tau \in I_x$. If*

$$[\mathbb{U}_\tau \psi_\lambda](x) = \psi_\lambda(\mathbf{s}_\tau(x)) = e^{\lambda \tau} \psi_\lambda(x), \quad \forall \tau \in I_x. \tag{7}$$

Then $\psi_\lambda(x)$ is called the open eigenfunction of the Koopman operator family $\mathbb{U}_t$, for $t \in \mathbb{R}$ with eigenvalue λ.

If $\boldsymbol{\mathcal{C}}$ is a proper invariant subset of $\mathcal{Z}$ in which case $I_x = \mathbb{R}$ for every $x \in \boldsymbol{\mathcal{C}}$, then ψ_λ is called the subdomain eigenfunction. If $\boldsymbol{\mathcal{C}} = \mathcal{Z}$ then ψ_λ will be the ordinary eigenfunction associated with eigenvalue λ as defined in (5). The open eigenfunctions as defined above can be extended from $\boldsymbol{\mathcal{C}}$ to a larger reachable set when $\boldsymbol{\mathcal{C}}$ is open based on the construction procedure outlined in [60, Definition 5.2, Lemma 5.1]. Let $\mathcal{P}$ be that larger domain. The eigenvalues of the linearization of the system dynamics at the origin, i.e., $E := \frac{\partial f}{\partial x}(0)$, will form the eigenvalues of the Koopman operator [60, Proposition 5.8]. Our interest will be in constructing the corresponding eigenfunctions, defined over the domain $\mathcal{P}$. We will refer to these eigenfunctions as *principal eigenfunctions* [60].
The principal eigenfunctions can be used as a change of coordinates in the linear representation of a nonlinear system and draw a connection to the famous Hartman-Grobman theorem on linearization and Poincare normal form [64]. The principal eigenfunctions will be defined over a proper subset $\mathcal{P}$ of the state space $\mathcal{Z}$ (called subdomain eigenfunctions) or over the entire $\mathcal{Z}$ [60, Lemma 5.1, Corollary 5.1, 5.2, and 5.8].

2.2 Decomposition of Koopman Eigenfunctions

Following [26], we decompose principle eigenfunctions into linear and nonlinear parts. Specifically, consider the decomposition of the nonlinear system into linear and nonlinear parts as

$$\dot{x} = f(x) = Ex + (f(x) - Ex) =: Ex + G(x). \tag{8}$$

where $E = \frac{\partial f}{\partial x}(0)$ with Ex the linear part and G the purely nonlinear part. For the simplicity of presentation and continuity of notations, we present approximation results for eigenfunctions with simple real eigenvalues; the extension to the complex case is deferred to future work. Let λ be the eigenvalues of the Koopman generator and also of E. The eigenfunction corresponding to eigenvalue λ admits the decomposition into linear and nonlinear parts.

$$\phi_\lambda(x) = w^\top x + h(x), \tag{9}$$

where $w^\top x$ and $h(x)$ are the eigenfunction's linear and purely nonlinear parts, respectively. Substituting (9) in following general expression of Koopman eigenfunction i.e.,

$$\frac{\partial \phi_\lambda}{\partial x}(x) \cdot f(x) = \lambda \phi_\lambda(x) \tag{10}$$

and using (8), we obtain following equations to be satisfied by w and $h(x)$

$$w^\top E = \lambda w^\top, \;\; \frac{\partial h}{\partial x}(x) \cdot f(x) - \lambda h(x) + w^\top G(x) = 0. \tag{11}$$

So, the linear part of the eigenfunction can be found as the left eigenvector with eigenvalue λ of the matrix E, and the nonlinear term satisfies the linear partial differential equation.

3 RKHS-based Computational Framework

3.1 Reproducing Kernel Hilbert Spaces (RKHS)

We give a brief overview of reproducing kernel Hilbert spaces as used in statistical learning theory [35]. Early work developing the theory of RKHS was undertaken by N. Aronszajn [65].

Definition 4. *Let $\mathcal{H}$ be a Hilbert space of functions on a set $\mathcal{X}$. Denote by $\langle f, g\rangle$ the inner product on $\mathcal{H}$ and let $\|f\| = \langle f, f\rangle^{1/2}$ be the norm in $\mathcal{H}$, for f and $g \in \mathcal{H}$. We say that $\mathcal{H}$ is a reproducing kernel Hilbert space (RKHS) if there exists a function $K : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$ such that*

i. $K_x := K(x, \cdot) \in \mathcal{H}$ *for all* $x \in \mathcal{H}$.

ii. K *spans* $\mathcal{H}$: $\mathcal{H} = \overline{\mathit{span}\{K_x \mid x \in \mathcal{X}\}}$.

iii. K *has the* reproducing property: $\forall f \in \mathcal{H},\ f(x) = \langle f, K_x\rangle$.

k will be called a reproducing kernel of $\mathcal{H}$. $\mathcal{H}_K$ will denote the RKHS $\mathcal{H}$ with reproducing kernel K where it is convenient to explicitly note this dependence.

The important properties of reproducing kernels are summarized in the following proposition.

Proposition 1. *[65] If K is a reproducing kernel of a Hilbert space $\mathcal{H}$, then*

i. $K(x, y)$ *is unique.*

ii. $\forall x, y \in \mathcal{X}$, $K(x, y) = K(y, x)$ *(symmetry).*

iii. $\sum_{i,j=1}^{q} \alpha_i \alpha_j K(x_i, x_j) \geq 0$ *for* $\alpha_i \in \mathbb{R}$, $x_i \in \mathcal{X}$ *and* $q \in \mathbb{N}_+$ *(positive definiteness).*

iv. $\langle K(x, \cdot), K(y, \cdot)\rangle = K(x, y)$.

Common examples of reproducing kernels defined on a compact domain $\mathcal{X} \subset \mathrm{R}^n$ are the (1) constant kernel: $K(x, y) = m > 0$ (2) linear kernel: $K(x, y) = x \cdot y$ (3) polynomial kernel: $K(x, y) = (1 + x \cdot y)^d$ for $d \in \mathbb{N}_+$ (4) Laplace kernel: $K(x, y) = e^{-||x-y||_2/\sigma^2}$, with $\sigma > 0$ (5) Gaussian kernel: $K(x, y) = e^{-||x-y||_2^2/\sigma^2}$, with $\sigma > 0$ (6) triangular kernel: $K(x, y) = \max\{0, 1 - \frac{||x-y||_2^2}{\sigma}\}$, with $\sigma > 0$. (7) locally periodic kernel: $K(x, y) = \sigma^2 e^{-2\frac{\sin^2(\pi||x-y||_2/p)}{\ell^2}} e^{-\frac{||x-y||_2^2}{2\ell^2}}$, with $\sigma, \ell, p > 0$.

Theorem 1. *[65] Let $K : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$ be a symmetric and positive definite function. Then there exists a Hilbert space of functions $\mathcal{H}$ defined on $\mathcal{X}$ admitting K as a reproducing Kernel. Conversely, let $\mathcal{H}$ be a Hilbert space of functions $f : \mathcal{X} \to \mathbb{R}$ satisfying $\forall x \in \mathcal{X}, \exists \kappa_x > 0$, such that $|f(x)| \leq \kappa_x \|f\|_{\mathcal{H}}, \quad \forall f \in \mathcal{H}$. Then $\mathcal{H}$ has a reproducing kernel K.*

Theorem 2. *[65] Let $K(x, y)$ be a positive definite kernel on a compact domain or a manifold X. Then there exists a Hilbert space $\mathcal{F}$ and a function $\Phi : X \to \mathcal{F}$ such that*

$$K(x, y) = \langle \Phi(x), \Phi(y) \rangle_{\mathcal{F}} \quad \textit{for} \quad x, y \in X.$$

Φ is called a feature map, and $\mathcal{F}$ a feature space[1].

Theorem 3. *[66] Let $x \in \mathbb{R}^d$, $u(x)$ be an unknown function and $\tilde{\phi} := (\tilde{\phi}_1, \ldots, \tilde{\phi}_M)$ be a collection of bounded linear functionals acting on u. We introduce the notation $Y \in \mathbb{R}^M$ as a vector whose entries are $Y_i = u(\tilde{\phi}_i) := \int_{\mathbb{R}^d} \tilde{\phi}_i(x) u(x) dx$. Then the optimal recovery of u given Y is the function u^* in $\mathcal{H}_K$ that solves the following optimization problem:*

$$\begin{aligned} \min_{u^* \in \mathcal{H}_K} \quad & ||u^*||^2_{\mathcal{H}_K} \\ s.t. \quad & u^*(\tilde{\phi}_i) = Y_i, \ 1, \ldots, M \end{aligned} . \tag{12}$$

The problem has a closed-form solution in the form

$$u^*(x) = K(x, \tilde{\phi}) K(\tilde{\phi}, \tilde{\phi})^{-1} Y, \tag{13}$$

where $K(x, \tilde{\phi})$ is a vector of length M with entries

$$K(x, \tilde{\phi})_i := \int_{\mathbb{R}^d} K(x, x') \tilde{\phi}_i(x') dx', \tag{14}$$

and $K(\tilde{\phi}, \tilde{\phi})$ is a $M \times M$ matrix with entries

$$K(\tilde{\phi}_i, \tilde{\phi}_j) := \int_{\mathbb{R}^{2d}} \tilde{\phi}_i(x) K(x, x') \tilde{\phi}_j(x') dx dx'. \tag{15}$$

3.2 Solving the linear PDE from data

Solving the linear PDE (11) can be framed as a quadratic optimization problem which can be solved by kernel regression. For each eigenvalue λ_k, $k = 1, \ldots, n$, define the best Gaussian approximation of h_{λ_k}, where h_{λ_k} is a function of d variables, to be the $h^*_{\lambda_k}$ that satisfies

[1]The dimension of the feature space can be infinite, for example in the case of the Gaussian kernel.

$$
\begin{aligned}
\min_{h^*_{\lambda_k} \in \mathcal{H}_K} \quad & ||h^*_{\lambda_k}||^2_{\mathcal{H}_K} \\
\text{s.t.} \quad & h^*_{\lambda_k}(0) = 0 \\
& \frac{\partial}{\partial x_j} h^*_{\lambda_k}(0) = 0,\ j = 1, \ldots, d \\
& \frac{\partial h^*_{\lambda_k}}{\partial x}(x_i) \cdot f(x_i) - \lambda_k h^*_{\lambda_k}(x_i) = -w_k^T G(x_i),\ i = 1, \ldots, N,
\end{aligned} \tag{16}
$$

where $x_i := (x_{1,i}, x_{2,i}, \ldots, x_{d,i})$ denotes our collocation points.

Defining the functionals

$$
\begin{aligned}
\tilde{\phi}_1(x) =& \delta_0(x) \\
\tilde{\phi}_{1+i}(x) =& \delta_0(x) \cdot \frac{\partial}{\partial x_i},\ i = 1, \ldots, d \\
\tilde{\phi}_{1+d+i}(x) =& \sum_{j=1}^{d} f(x_i)_j \delta_{x_i}(x) \cdot \frac{\partial}{\partial x_j} - \lambda_k \delta_{x_i}(x),\ i = 1, \ldots, N
\end{aligned}, \tag{17}
$$

where $\delta_{x_i}(x)$ is the Dirac delta distribution centered at x_i, we solve the optimization problem (16) using a slight modification of the representer formula (13) at the end of Section 3.1:

$$
h^*_{\lambda_k}(x) = K(x, \tilde{\phi})(K(\tilde{\phi}, \tilde{\phi}) + \eta I)^{-1} Y, \tag{18}
$$

where $K(x, \tilde{\phi}) \in \mathbb{R}^{N+d+1}$ has entries

$$
K(x, \tilde{\phi})_i = \begin{cases} K(x, 0),\ i = 1 \\ \dfrac{\partial}{\partial x'} K(x, x')|_{x'=0},\ i = 2, \ldots, 1 + d \\ \displaystyle\sum_{j=1}^{d} f(x'_i) \frac{\partial}{\partial x'} K(x, x')|_{x'_i} - \lambda_k K(x, x'_i),\ i = 2 + d, \ldots, 1 + d + N \end{cases}. \tag{19}
$$

η is a small positive regularization constant that reduces numerical errors associated with inverting the matrix $K(\tilde{\phi}, \tilde{\phi})$, and $Y \in \mathbb{R}^{N+d+1}$ has entries $Y_1 = \cdots = Y_{d+1} = 0$ and $Y_{d+1+i} = -G(x_i)^T w_k,\ i = 1, \ldots, N$.

4 Error Estimates

Theorem 4 (Validity of Stability and Smoothness Assumptions)**.** *Let the equilibrium point x_e of the ODE* (1) *be hyperbolic. Assume that $f \in C^m(\Omega, \mathbb{R}^d)$ for some integer $m \geq 1$, where Ω is an open neighborhood around x_e.*

Let $w \in \mathbb{R}^d$ be a fixed nonzero weight vector. Consider the partial differential equation (PDE)

$$\nabla h(x) \cdot f(x) - \lambda h(x) = -w^\top G(x), \quad x \in \Omega, \tag{20}$$

with boundary conditions

$$h(x_e) = 0, \quad \nabla h(x_e) = 0. \tag{21}$$

We introduce the differential operator D to refer to the left hand side of the PDE 20: $(Dh)(x) := \nabla h(x) \cdot f(x) - \lambda h(x)$. Then, the solution h satisfies:

1. *$h \in W_2^{m+1}(\Omega)$, i.e., h belongs to the Sobolev space $W_2^{m+1}(\Omega)$.*
2. *There exists a stability bound of the form*

$$\|h\|_{L_p(\Omega)} \leq C_D \|Dh\|_{L_q(\Omega)} + C_0 |h(x_e)| + C_0' \|\nabla h(x_e)\|_{\ell^r}, \tag{22}$$

   where $C_D, C_0, C_0' > 0$ are constants, and $p, q, r \in [1, \infty]$.

Remark 1. *Let us explain the meaning of each term on the right-hand side of (22):*

- ***Term 1:** $\|Dh\|_{L^q(\Omega)}$ This term measures the discrepancy between h and the PDE constraint imposed by the Koopman eigenvalue equation. If h is an exact solution to the PDE, then $Dh = -w^\top G$, and this residual is small. Informally, it captures how well h satisfies the PDE over the domain. It is the main driver of the solution's structure and penalizes deviation from the dynamic constraints.*

- ***Term 2:** $|h(x_e)|$ This is simply the absolute value of the function h evaluated at a reference point $x_e \in \Omega$. It introduces a scalar constraint that helps control the behavior of h at a specific location. Informally, this term "anchors" the solution to avoid degenerate cases where h could be arbitrarily shifted by adding constants (which may lie in the null space of D). It helps fixing the vertical offset of h.*

- ***Term 3:** $\|\nabla h(x_e)\|_{\ell^r}$ The term $\|\nabla h(x_e)\|_{\ell^r}$ represents the ℓ^r-norm of the gradient vector of the function h evaluated at the point x_e. If the domain $\Omega \subset \mathbb{R}^d$, then the gradient $\nabla h(x_e)$ is the vector $\nabla h(x_e) = \left(\frac{\partial h}{\partial x_1}(x_e), \ldots, \frac{\partial h}{\partial x_d}(x_e)\right)$, and the norm is defined as $\|\nabla h(x_e)\|_{\ell^r} := \left(\sum_{i=1}^d \left|\frac{\partial h}{\partial x_i}(x_e)\right|^r\right)^{1/r}$, for $1 \leq r < \infty$, and $\|\nabla h(x_e)\|_{\ell^\infty} := \max_{1 \leq i \leq d} \left|\frac{\partial h}{\partial x_i}(x_e)\right|$.*

  Informally, this quantity measures the local rate of change (or "steepness") of h at the point x_e. It quantifies how rapidly h varies near that point, and plays an important role in the stability bound (22).

In the context of Equation (22), we are bounding the global behavior of h (in the L^p norm over the domain) by combining two types of information: i.) The residual Dh, which controls how closely h satisfies the PDE; ii.) Local "anchor" terms, $h(x_e)$ and $\nabla h(x_e)$, which ensure uniqueness and prevent degeneracy (e.g., the solution drifting within the nullspace of D).

The presence of $\|\nabla h(x_e)\|_{\ell^r}$ can be seen as providing a "pointwise grip" on the solution: even if Dh is small everywhere, the function h could still shift in flat directions unless it is tied down locally. This term helps enforce that control.

This term quantifies the magnitude of the gradient vector at a single point in terms of the standard ℓ^r-norm on $\mathbb{R}^d$. It is included in the stability estimate to ensure control over the local behavior of the solution h near the reference point x_e, particularly to eliminate degeneracies arising from the kernel of the differential operator.

Together, these terms provide a well-balanced estimate that ensures global control of h through its PDE residual and localized behavior. This is crucial in learning Koopman eigenfunctions where directly controlling the operator spectrum is challenging, and instead, we rely on function-level estimates like (22) to enforce stability and uniqueness. Finally, $h(x_e) = 0$ and $\nabla h(x_e) = 0$ from (21).

Proof. We first observe that the nonlinear term $G(x) = f(x) - Ex$, where $E = \frac{\partial f}{\partial x}\Big|_{x_e}$, has smoothness $G \in C^m(\Omega, \mathbb{R}^d)$.

Step 1: Injectivity of D

Since x_e is hyperbolic, the Jacobian matrix E has no eigenvalues with zero real parts. This implies that the linearized system around x_e has no neutral modes, and thus the operator D associated with the PDE has no non-trivial solutions to the homogeneous equation $Dh = 0$.

By the Hartman-Grobman theorem, the behavior of the nonlinear system near x_e is qualitatively similar to its linearization. Therefore, the vector field $f(x)$ can be approximated by $f(x) \approx E(x - x_e)$ near x_e, and the operator D reduces to a linear problem in this neighborhood. The only solution to $Dh = 0$ near x_e is the trivial solution $h(x) = 0$.

Thus, D is injective:

$$\ker(D) = \{h \in W_2^{m+1}(\Omega) \mid Dh = 0\} = \{0\}.$$

Step 2: Surjectivity of D

To prove surjectivity, consider the equation:

$$Dh(x) = -w^\top G(x),$$

where $G(x) \in C^m(\Omega)$. The solution to this equation is governed by regularity theory of Elliptic PDEs. Specifically, for smooth forcing terms $G(x)$, elliptic regularity guarantees that the operator D will have a solution in the Sobolev space $W_2^{m+1}(\Omega)$, i.e., $h(x) \in W_2^{m+1}(\Omega)$.

Elliptic regularity theory ensures that if the right-hand side $-w^\top G(x)$ is sufficiently smooth, the solution $h(x)$ will also be smooth and lie in $W_2^{m+1}(\Omega)$. Therefore, D is surjective, meaning that for any smooth $G(x)$, there exists a solution $h(x) \in W_2^{m+1}(\Omega)$.

Step 3: Bounded Invertibility via the Banach Inverse Mapping Theorem

Since D is a bounded linear operator (under the smoothness and boundedness assumptions on $f(x)$ and λ) that is both injective and surjective, it is a bounded linear bijection between $W_2^{m+1}(\Omega)$ and $L_q(\Omega)$.

By the Banach Inverse Mapping Theorem, the inverse operator D^{-1} exists and is bounded. Therefore, there exists a constant $C_D > 0$ such that:

$$\|h\|_{W_2^{m+1}(\Omega)} \leq C_D \|Dh\|_{L_q(\Omega)}$$

Step 4: Stability Bound for $h(x)$

The solution $h(x)$ lies in the Sobolev space $W_2^{m+1}(\Omega)$, and we apply Sobolev embedding theorems to obtain a bound for $h(x)$ in $L_p(\Omega)$. Sobolev embedding states that for an embedding $W_2^{m+1}(\Omega) \subset L_p(\Omega)$, the solution $h(x)$ will be in $L_p(\Omega)$ for some p, provided that m is large enough.

The stability bound is expressed as:

$$\|h\|_{L_p(\Omega)} \leq C_D \|Dh\|_{L_q(\Omega)} + C_0 |h(x_e)| + C_0' \|\nabla h(x_e)\|_{\ell^r}.$$

This bound reflects the control over $h(x)$ in terms of the forcing term $G(x)$ (through the L_q-norm of $Dh(x)$) and the boundary conditions $h(x_e)$ and $\nabla h(x_e)$.

□

To formulate the following convergence result we make use of the fill distance $\rho_{Z,\Omega}$ of the finite set $Z \subset \Omega$, given by

$$\rho_{Z,\Omega} := \sup_{x \in \Omega} \min_{z \in Z} \|x - z\|_2.$$

Theorem 5. *Assume that Theorem 4 holds with constants $m > d/2$, $p, q \in [1, \infty]$, $C_D > 0$, and let h_λ be the solution of the PDE* (20)*. Assume furthermore that $\mathcal{H} \hookrightarrow W_2^{m+1}(\Omega)$, and let h_λ^* be the solution of the corresponding optimization problem* (16) *with collocation points $Z \subset \Omega$ and with no regularization (i.e., $\eta = 0$ in* (18)*).*

Then there are constants $\rho_0, C > 0$ depending on d, m, Ω, p, q, but not on λ, f, Z, h_λ, such that if $\rho_{Z,\Omega} < \rho_0$ it holds

$$\|h_\lambda - h_\lambda^*\|_{L_p(\Omega)} \leq C \left(\|f\|_{W_2^m(\Omega, \mathbb{R}^d)} + |\lambda| \right) \rho_{Z,\Omega}^{m - d\left(\frac{1}{2} - \frac{1}{q}\right)_+} \|h_\lambda\|_{\mathcal{H}},$$

where $(x)_+ := \max(x, 0)$.

Proof. We first show that $h \in W_2^{m+1}(\Omega)$ implies that $Dh \in W_2^m(\Omega)$, and give an estimate on its norm. Namely, for any $\alpha \in \mathbb{N}_0^d$ with $|\alpha| \leq m$ we have

$$\begin{aligned}\partial^\alpha Dh(z) &= \partial^\alpha \left(f(z)^T \nabla h(z) - \lambda h(z)\right)\\ &= \sum_{i=1}^d \left(\partial^\alpha f_i(z)\right) (\nabla h(z))_i + \sum_{i=1}^d f_i(z) \partial^\alpha (\nabla h(z))_i - \lambda \partial^\alpha h(z),\end{aligned}$$

and thus there is a constant C_d depending on d such that

$$\begin{aligned}\|\partial^\alpha Dh(z)\|_{L_2} &\leq d \max_{1\leq i\leq d} \left(\|\partial^\alpha f_i\|_{L_2} \|(\nabla h)_i\|_{L_2} + \|f_i\|_{L_2} \|\partial^\alpha (\nabla h)_i\|_{L_2}\right) + |\lambda| \|\partial^\alpha h\|_{L_2}\\ &\leq C_d \left(\|f\|_{W_2^m(\Omega,\mathbb{R}^d)} + |\lambda|\right) \|h\|_{W_2^{m+1}}.\end{aligned}$$

This in turns implies that for some constant $C_{d,m} > 0$ depending on d and m we have

$$\|Dh\|_{W_2^m} = \left(\sum_{|\alpha|\leq m} \|\partial^\alpha Dh\|_{L_2}^2\right)^{1/2} \leq C_{d,m} \left(\|f\|_{W_2^m(\Omega,\mathbb{R}^d)} + |\lambda|\right) \|h\|_{W_2^{m+1}}. \quad (23)$$

This proves in particular that $Dh_\lambda^* \in W_2^m(\Omega)$ since $h_\lambda^* \in \mathcal{H} \hookrightarrow W_2^{m+1}$, while $Dh_\lambda \in W_2^{m+1}(\Omega)$ by assumption and by Theorem 4.

Since $m > d/2$, we may now use the zero lemma of Theorem 12 in [67], stating that there are $\rho_0, C > 0$, depending on Ω, p, q, such that if $\rho_{Z,\Omega} < \rho_0$ it holds

$$\|Dh_\lambda - Dh_\lambda^*\|_{L_q(\Omega)} \leq C \rho_{Z,\Omega}^{m-d\left(\frac{1}{2}-\frac{1}{q}\right)_+} \|Dh_\lambda - Dh_\lambda^*\|_{W_2^m(\Omega)}, \quad (24)$$

and using the linearity of D and the bound (23) this gives

$$\|Dh_\lambda - Dh_\lambda^*\|_{L_q} \leq C C_{d,m} \left(\|f\|_{W_2^m(\Omega,\mathbb{R}^d)} + |\lambda|\right) \rho_{Z,\Omega}^{m-d\left(\frac{1}{2}-\frac{1}{q}\right)_+} \|h_\lambda - h_\lambda^*\|_{W_2^{m+1}}. \quad (25)$$

We conclude by bounding the norm in the right-hand side. Since h_λ satisfies the constraints of the problem (16), and h_λ^* is its minimal norm solution, we have $\|h_\lambda^*\|_{\mathcal{H}} \leq \|h_\lambda\|_{\mathcal{H}}$. Furthermore, by assumption there is $C_e > 0$ such that $\|h\|_{W_2^{m+1}} \leq C_e \|h\|_{\mathcal{H}}$ for all $h \in \mathcal{H}$. It follows that

$$\|h_\lambda - h_\lambda^*\|_{W_2^{m+1}} \leq C_e \|h_\lambda - h_\lambda^*\|_K \leq C_e \left(\|h_\lambda\|_{\mathcal{H}} + \|h_\lambda^*\|_K\right) \leq 2C_e \|h_\lambda\|_{\mathcal{H}},$$

and thus (25) simplifies to

$$\|Dh_\lambda - Dh_\lambda^*\|_{L_q} \leq 2C C_{d,m} C_e \left(\|f\|_{W_2^m(\Omega,\mathbb{R}^d)} + |\lambda|\right) \rho_{Z,\Omega}^{m-d\left(\frac{1}{2}-\frac{1}{q}\right)_+} \|h_\lambda\|_{\mathcal{H}}. \quad (26)$$

We can now define $u := h_\lambda - h_\lambda^*$ and observe that $u \in W_2^{m+1}$, since $h_\lambda \in W_2^m$ by assumption, and since we proved that $h_\lambda^* \in W_2^{m+1}$. We may now insert u in the stability bound (22), and since $Du = Dh_\lambda - Dh_\lambda^*$ we may further bound the right-hand side by (26). This gives the result with an appropriate constant $C > 0$, since both h_λ and h_λ^* satisfy the boundary conditions (21). □

Remark 2. *We point out that in Theorem 5 the factor* $(1/2 - 1/q)_+$ *is decreasing in* q *and vanishes for* $1 \leq q \leq 2$. *It follows that, given a target* p, *one should choose the smallest* q *such that* (22) *holds.*

Remark 3. *Observe that the assumption* $\mathcal{H} \hookrightarrow W_2^{m+1}(\Omega)$ *is satisfied whenever the kernel* K *has smoothness* $C^{m+1}(\Omega)$ *in both variables separately and* $\Omega \subset \mathbb{R}^d$ *has a sufficiently smooth boundary (see e.g. Chapter 1 in [68], or Chapter 10 in [69]). Many kernels of common use meet this condition, such as the Wendland and Matèrn kernels of suitable regularity (see [69]), or the Gaussian kernel (see [70]).*

Moreover, for $W_2^m(\Omega)$ *to be a RKHS one needs* $m > d/2$, *while the additional order* $m+1$ *is required in order to compute the partial derivatives in the method of Section 3.2. Apart from these restrictions, the actual value of* m *can be chosen to maximize the convergence speed in Theorem 5, where the maximal* m *is determined by the smoothness of* f, *which is a datum of the problem.*

5 Examples

5.1 First Analytical Example

Consider the following dynamical system with an equilibrium point at the origin.

$$\dot{x} = \begin{bmatrix} -2\lambda_2 x_2(x_1^2 - x_2 - 2x_1x_2^2 + x_2^4) + \lambda_1(x_1 + 4x_1^2x_2 - x_2^2 - 8x_1x_2^3 + 4x_2^5) \\ 2\lambda_1(x_1 - x_2^2)^2 - \lambda_2(x_1^2 - x_2 - 2x_1x_2^2 + x_2^4) \end{bmatrix}. \tag{27}$$

The eigenvalues of the linearization of the system at the origin i.e, E are $\lambda_1 = -1$ and $\lambda_2 = 3$. For this example, the principal eigenfunctions can be computed analytically and are as follows:

$$\phi_{\lambda_1}(x) = x_1 - x_2^2, \quad \lambda_1 = -1 \text{ and}$$
$$\phi_{\lambda_2}(x) = -x_1^2 + x_2 + 2x_1x_2^2 - x_2^4, \quad \lambda_2 = 3.$$

Using 3600 points over the square $[-1, 1] \times [-1, 1]$, we apply kernel regression with the 2D Gaussian kernel

$$K(x_1, x_2; \sigma, \sigma_2) = \exp\Big(- \frac{(x_1 - y_1)^2}{2\sigma_1^2} - \frac{(x_2 - y_2)^2}{2\sigma_2^2}\Big) \tag{28}$$

to learn the eigenfunction ϕ_{λ_k} and depict the results in (1) and (2). For ϕ_{λ_1}, we take $\sigma_1 = \sigma_2 = 2$; for ϕ_{λ_2}, we use $\sigma_1 = 2$ and $\sigma_2 = 3$. The pointwise relative error is very low with the exception of the points where the true solution is equal to zero.

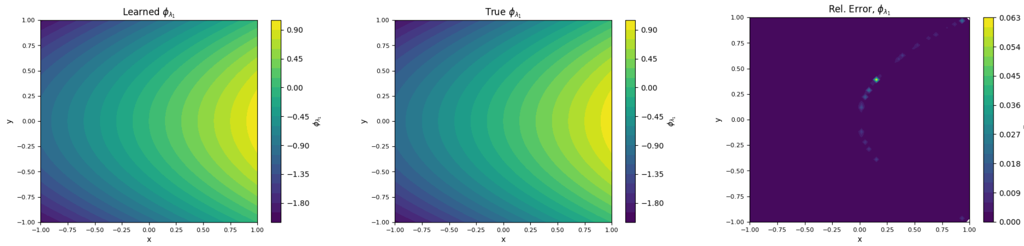

Figure 1: Learned $\phi^*_{\lambda_1}$ (left), true ϕ_{λ_1} (center) and relative error (right)

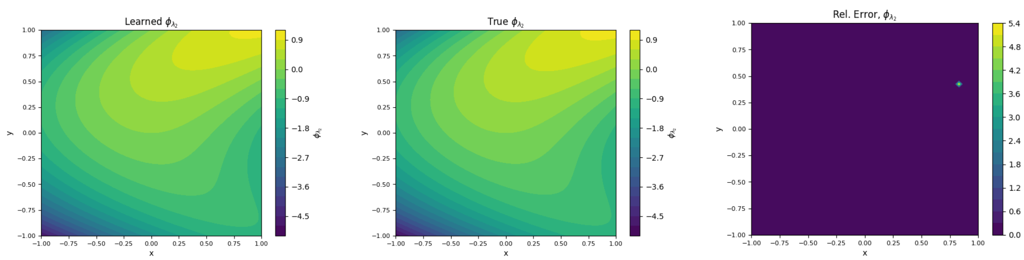

Figure 2: Learned $\phi^*_{\lambda_2}$ (left), true ϕ_{λ_2} (center) and relative error (right)

5.2 Second Analytical Example

$$\dot{x} = \begin{bmatrix} \frac{\left(7.5x_2^2+5.0\right)\left(x_1^3+x_1+\sin(x_2)\right)+\left(-x_1+x_2^3+2x_2\right)\cos(x_2)}{9x_1^2x_2^2+6x_1^2+3x_2^2+\cos(x_2)+2} \\ \\ \frac{2.5x_1^3+2.5x_1-\left(3x_1^2+1\right)\left(-x_1+x_2^3+2x_2\right)+2.5\sin(x_2)}{9x_1^2x_2^2+6x_1^2+3x_2^2+\cos(x_2)+2} \end{bmatrix}. \tag{29}$$

The vector field for this system is smooth everywhere on $\mathbb{R}^2$, with a saddle point at the origin. One can verify that the two principal eigenpairs of this system are given by

$$\phi_{\lambda_1}(x) = x_1 - 2x_2 - x_2^3, \quad \lambda_1 = -1 \text{ and}$$
$$\phi_{\lambda_2}(x) = x_1 + \sin(x_2) + x_1^3, \quad \lambda_2 = 2.5.$$

We apply the method outlined in Section 3.2 and use the 2D Gaussian kernel to solve the PDE for each eigenvalue over 2500 points in the grid $[1.5, 2.5] \times [1.5, 2.5]$,

For λ_1, we set $\sigma_1 = \sigma_2 = 3$ and for λ_2, we use $\sigma_1 = \sigma_2 = 7$. As shown by Figures 3 and 4, the solution $\phi^*_{\lambda_k}$, where the nonlinear part $h^*_{\lambda_l}$ has been recovered by the Gaussian kernel, is an accurate approximation of the true eigenfunction ϕ_{λ_k} for both eigenvalues λ_k.

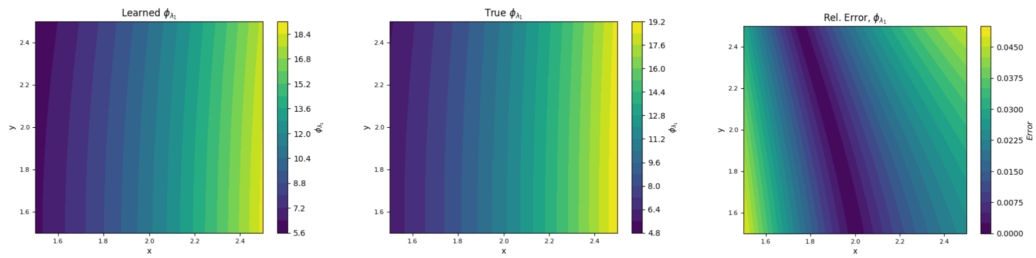

Figure 3: Learned $\phi^*_{\lambda_1}$ (left), true ϕ_{λ_1} (center) and relative error (right)

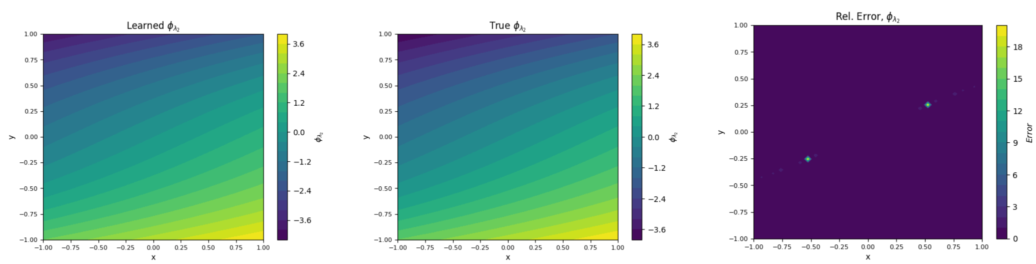

Figure 4: Learned $\phi^*_{\lambda_2}$ (left), true ϕ_{λ_2} (center) and relative error (right)

5.3 The Duffing Oscillator

Consider the unforced Duffing oscillator, described by

$$\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= -\delta x_2 - x_1(\beta + \alpha x_1^2) \end{aligned} \tag{30}$$

with $\delta = 0.5$, $\beta = -1$, and $\alpha = 1$, where $z \in \mathbb{R}$ and $\dot{z} \in \mathbb{R}$ are the scalar position and velocity, respectively. The dynamics admit two stable equilibrium points at $(-1, 0)$ and $(1, 0)$, and one unstable equilibrium point at the origin. In this example, we sample 2500 points over the domain $[-2, 2] \times [-2, 2]$ and use the 2D Gaussian kernel with $\sigma_1 = \sigma_2 = 15$. Our results in Figure 5 depict $\phi^*_{\lambda_1}$ for $\lambda_1 = \frac{-1+\sqrt{17}}{4}$; this time, we consider only the larger eigenvalue. The plot accurately captures the behavior of the eigenfunction of the Koopman operator corresponding to λ_1 up to scaling.

5.4 A Three Dimensional Gradient System

Consider a three-dimensional gradient system of the form

$$\dot{x} = -\frac{\partial V}{\partial x}, \quad x = (x_1\ x_2\ x_3)^\top, \tag{31}$$

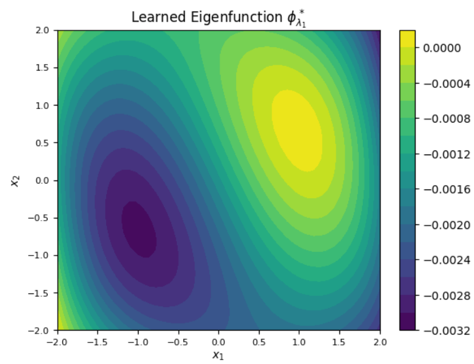

Figure 5: Learned $\phi^*_{\lambda_1}$

where the potential function V is given by

$$V(x) = x^\top P x + e^{-(x_1 - x_2)^2}, \tag{32}$$

where $P = \begin{pmatrix} 0.2 & 0.1 & 0.05 \\ 0.1 & 0.3 & 0.05 \\ 0.05 & 0.05 & 0.2 \end{pmatrix}$ is a positive definite matrix. The system admits an unstable equilibrium at the origin and two stable minima at $x = (0.90, -0.73, 0.14)$ and $x = (-0.90, 0.73, 0.14)$. We first compute Jacobian matrix of $-\frac{\partial V}{\partial x}$ at $x = 0$ which has three real eigenvalues, i.e., 3.70, −0.29, and −0.81. To construct $\phi_\lambda(x)$, we solve (16) over the domain $[-2, 2] \times [-2, 2] \times [-2, 2]$ by sampling 3379 points and use the Gaussian kernel $K(x_1, x_2) = \exp\left(-\frac{(x_1 - x_2)^2}{2 \times 1.1^2}\right)$. To visualize the result, we plot the level curve of $\phi_\lambda(x)$ with the unstable eigenvalue $\lambda = 3.70$ with a fixed $z = 0.57$. As shown in Figure 6, the level curve reveals two regions of attraction centered at the local minima of $V(x)$, i.e., x_1, x_2.

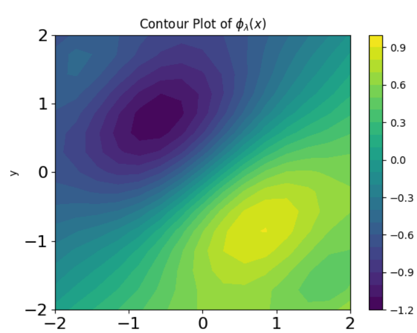

Figure 6: Learned $\phi^*_{\lambda_1}$

6 Code

The code for this paper can be found at https://github.com/jonghyeon1998/Koopman

7 Discussion

7.1 Training kernel parameters

Although we have focused on 2D and 3D examples, in practice, many ODEs with real-life applications have a high-dimensional state space. High-dimensional problems pose challenges due to the large number of collocation points required to capture the behavior of the solution. In particular, performing large kernel matrix computations becomes very computationally expensive with $O(N^3)$ time complexity in the number of collocation points. Another challenge associated with high dimensionality is the fact that the kernel parameters typically have to be well-tuned to the data to produce accurate kernel regressors. We offer a method which is a deterministic variant of Kernel Flows (KF) [71], a well-established cross-validation algorithm for learning kernel parameters that has been successfully applied to various dynamical systems [53, 72, 73, 43, 52, 74, 75]. Given a kernel K_θ dependent on parameters θ, the idea is to find the minimizer of the following loss function:

$$\rho(\theta) = \frac{1}{N}\sum_{j=1}^{N} 1 - \frac{Y^{j,T}(K_\theta^j(\tilde{\phi},\tilde{\phi}) + \eta I)^{-1}Y^j}{Y^T(K_\theta(\tilde{\phi},\tilde{\phi}) + \eta I)^{-1}Y}. \tag{33}$$

The matrix $K_\theta^{-j}(\tilde{\phi},\tilde{\phi}) \in \mathbb{R}^{(N+d)\times(N+d)}$ is formed by removing the $1+d+j$th row and column of $K_\theta(\tilde{\phi},\tilde{\phi}) \in \mathbb{R}^{(N+d+1)\times(N+d+1)}$, which correspond to the jth data point $x_j \in \mathbb{R}^d$, and $Y^{-j} = (Y_1, \ldots, Y_{1+d}, \ldots, Y_{1+d+j-1}, Y_{1+d+j+1}, \ldots, Y_{1+N+d})^T \in \mathbb{R}^{N+d}$ is likewise constructed by removing the $1+d+j$th entry of Y. The intuition behind the function $\rho(\theta)$ is that the kernel parameters are considered good if the interpolant $u^*(x) = K(x,\tilde{\phi})K(\tilde{\phi},\tilde{\phi})^{-1}Y$ does not change much when we remove a single collocation point.

7.2 Repeated and complex eigenvalues

The method in our paper can be generalized to dynamical systems where the eigenvalues of E are complex or repeated.

7.2.1 Repeated Eigenvalues

We now assume λ is an eigenvalue of E with algebraic multiplicity two and geometric multiplicity one. Then we have generalized eigenvectors, w_1, w_2 for eigenvalues λ as

$$w_1^\top E = \lambda w_1^\top, \quad w_2^\top E = \lambda w_2^\top + w_1^\top. \tag{34}$$

Similarly, the generalized eigenfunctions $\phi_1(x)$ and $\phi_2(x)$ are of the form

$$\phi_1(x) = w_1^\top x + h_1(x), \quad \phi_2(x) = w_2^\top x + h_2(x). \tag{35}$$

In particular, $\phi_1(x)$,$\phi_2(x)$ satisfies

$$\frac{\partial \phi_1}{\partial x}(x) \cdot f(x) = \lambda \phi_1(x), \quad \frac{\partial \phi_2}{\partial x}(x) \cdot f(x) = \lambda \phi_2(x) + \phi_1(x), \tag{36}$$

where $h_1(x)$ and $h_2(x)$ satisfy

$$\begin{aligned} &\frac{\partial h_1}{\partial x}(x) \cdot f(x) - \lambda h_1(x) + w_1^\top G(x) = 0, \\ &(w_2^\top + \frac{\partial h_2}{\partial x}(x))(Ex + G(x)) = \lambda(w_2^\top x + h_2(x)) + w_1^\top x + h_1(x). \end{aligned} \tag{37}$$

Since $w_2^\top E = \lambda w_2^\top + w_1^\top$, we have

$$\begin{aligned} \frac{\partial h_1}{\partial x}(x) \cdot f(x) - \lambda h_1(x) + w_1^\top G(x) = 0. \\ \frac{\partial h_2}{\partial x}(x) \cdot f(x) - \lambda h_2(x) - h_1(x) + w_2^\top G(x) = 0. \end{aligned} \tag{38}$$

Written in matrix form, we have

$$\begin{pmatrix} \frac{\partial h_1}{\partial x}(x) \\ \frac{\partial h_2}{\partial x}(x) \end{pmatrix} + \begin{pmatrix} w_1^\top G(x) \\ w_2^\top G(x) \end{pmatrix} - \begin{pmatrix} \lambda & 0 \\ 1 & \lambda \end{pmatrix} \begin{pmatrix} h_1(x) \\ h_2(x) \end{pmatrix} = 0. \tag{39}$$

That is,

$$\begin{pmatrix} \frac{\partial h_1}{\partial x}(x) \\ \frac{\partial h_1}{\partial x}(x) \end{pmatrix} + W^\top G(x) - \Lambda \begin{pmatrix} h_1(x) \\ h_2(x) \end{pmatrix} = 0. \tag{40}$$

The eigenfunctions $\phi_1(x), \phi_2(x)$ can be learned by first solving for $h_1(x)$ with the generalized representer theorem in Section 3 and using our recovered $h_1(x)$ to solve for $h_2(x)$.

7.2.2 Complex Eigenvalues

We now consider eigenvalues of the form $\lambda = \lambda_{\text{Re}} + j\lambda_{\text{Im}}$. The corresponding eigenvector is given by

$$(w_{\text{Re}}^\top + jw_{\text{Im}}^\top)E = (\lambda_{\text{Re}} + j\lambda_{\text{Im}})(w_{\text{Re}}^\top + jw_{\text{Im}}^\top) = \lambda_{\text{Re}} w_{\text{Re}}^\top - \lambda_{\text{Im}} w_{\text{Im}}^\top + j(\lambda_{\text{Im}} w_{\text{Re}}^\top + \lambda_{\text{Re}} w_{\text{Im}}^\top). \tag{41}$$

Hence, we have

$$w_{\text{Re}}^\top E = \lambda_{\text{Re}} w_{\text{Re}}^\top - \lambda_{\text{Im}} w_{\text{Im}}^\top, \quad w_{\text{Im}}^\top E = \lambda_{\text{Im}} w_{\text{Re}}^\top + \lambda_{\text{Re}} w_{\text{Im}}^\top \quad . \tag{42}$$

And the eigenfunction can be written as

$$\phi(x) = \phi_{\mathrm{Re}}(x) + j\phi_{\mathrm{Im}}(x) = w_{\mathrm{Re}}^\top x + h_{\mathrm{Re}}(x) + j(w_{\mathrm{Im}}^\top x + h_{\mathrm{Im}}(x)), \tag{43}$$

where $h_{\mathrm{Re}}(x)$ and $h_{\mathrm{Im}}(x)$ satisfy

$$\left(w_{\mathrm{Re}}^\top + jw_{\mathrm{Im}}^\top + \frac{\partial h_{\mathrm{Re}}}{\partial x}(x) + j\frac{\partial h_{\mathrm{Im}}}{\partial x}(x)\right)(Ex + G(x)) = (\lambda_{\mathrm{Re}} + j\lambda_{\mathrm{Im}})(w_{\mathrm{Re}}^\top x + h_{\mathrm{Re}}(x) + j(w_{\mathrm{Im}}^\top x + h_{\mathrm{Im}}(x))). \tag{44}$$

That is,

$$\begin{aligned}
&(w_{\mathrm{Re}}^\top + jw_{\mathrm{Im}}^\top)Ex = (\lambda_{\mathrm{Re}} + j\lambda_{\mathrm{Im}})(w_{\mathrm{Re}}^\top + jw_{\mathrm{Im}}^\top)x,\\
&(\frac{\partial h_{\mathrm{Re}}}{\partial x}(x) + j\frac{\partial h_{\mathrm{Im}}}{\partial x}(x))f(x) + (w_{\mathrm{Re}}^\top + jw_{\mathrm{Im}}^\top)G(x) = (\lambda_{\mathrm{Re}} + j\lambda_{\mathrm{Im}})(h_{\mathrm{Re}}(x) + jh_{\mathrm{Im}}(x)).
\end{aligned} \tag{45}$$

Therefore, we have

$$\begin{array}{c}
\frac{\partial h_{\mathrm{Re}}}{\partial x}(x)\cdot f(x) + w_{\mathrm{Re}}^\top G(x) = \lambda_{\mathrm{Re}}h_{\mathrm{Re}}(x) - \lambda_{\mathrm{Im}}h_{\mathrm{Im}}(x)\\
\frac{\partial h_{\mathrm{Im}}}{\partial x}(x)\cdot f(x) + w_{\mathrm{Im}}^\top G(x) = \lambda_{\mathrm{Im}}h_{\mathrm{Re}}(x) + \lambda_{\mathrm{Re}}h_{\mathrm{Im}}(x)
\end{array}. \tag{46}$$

Defining $h(x) := (h_{\mathrm{Re}}(x), h_{\mathrm{Im}}(x))^\top \in \mathbb{R}^2$ and $\frac{\partial h}{\partial x}(x) \in \mathbb{R}^{2\times d}$ is a matrix with entries $(\frac{\partial h}{\partial x}(x))_{ij} = \frac{\partial h_i}{\partial x_j}(x)$, we can still rewrite (46) in matrix form:

$$\begin{pmatrix}\frac{\partial h_{\mathrm{Re}}}{\partial x}(x)\cdot f(x)\\ \frac{\partial h_{\mathrm{Im}}}{\partial x}(x)\cdot f(x)\end{pmatrix} + \begin{pmatrix}w_{\mathrm{Re}}^\top G(x)\\ w_{\mathrm{Im}}^\top G(x)\end{pmatrix} - \begin{pmatrix}\lambda_{\mathrm{Re}} & -\lambda_{\mathrm{Im}}\\ \lambda_{\mathrm{Im}} & \lambda_{\mathrm{Re}}\end{pmatrix}\begin{pmatrix}h_{\mathrm{Re}}(x)\\ h_{\mathrm{Im}}(x)\end{pmatrix} = 0. \tag{47}$$

That is,

$$\frac{\partial h}{\partial x}(x)\cdot f(x) + W^\top G(x) - \Lambda h(x) = 0. \tag{48}$$

Then the optimal recovery h^* of h admits the following representer formula:

$$h^*(x) = K(x, \tilde{\phi})K(\tilde{\phi}, \tilde{\phi})^{-1}Y, \tag{49}$$

using the notation introduced in Section 3, where $K(x, \tilde{\phi}) \in \mathbb{R}^{2\times 2(M+1+d)}$, $K(\tilde{\phi}, \tilde{\phi}) \in \mathbb{R}^{2(M+1+d)\times 2(M+1+d)}$, $Y \in \mathbb{R}^{2(M+1+d)}$; see [76] for more details on matrix-valued kernels.

8 Conclusion

In this paper, we have introduced a novel kernel-based method for approximating the principal eigenfunctions of the Koopman operator, offering a computationally efficient alternative to directly calculating the operator itself. By leveraging the decomposition of the eigenfunctions into linear and nonlinear components, we have shown that the linear part corresponds to the local linear dynamics of the system, while the nonlinear part can be obtained through kernel methods.

Our approach not only makes the computation of Koopman eigenfunctions more tractable but also enhances the ability to analyze complex dynamical systems by providing valuable insights into their long-term behavior. Overall, the kernel-based method for the construction of eigenfunctions of the Koopman operator presented here represents a significant step toward bridging the gap between linear and nonlinear dynamics within a rigorous mathematical framework.

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